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Chirality-dependent interfacial energy dissipation in graphene-reinforced polymer nanocomposites: A molecular dynamics study

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ABSTRACT

Methods for controlling the macroscopic mechanical behavior of polymers through graphene insertion are now being refined to utilize the inherent chirality of graphene. In this study, we computationally investigated the variation in the damping capacity with respect to graphene chirality and the relative vibrational load direction in epoxy/graphene nanocomposites at the molecular scale. All-atom molecular dynamics simulations showed that the nanocomposites exhibited improved energy dissipation when subjected to out-of-plane oscillatory shear strain along the armchair graphene compared to that under shear strain along the zigzag graphene. In particular, the anisotropic behavior governed the interfacial friction and associated slip properties of the polymer adjacent to graphene. Due to its high elastic modulus, armchair graphene facilitated the slippage of polymer components by suppressing out-of-plane wrinkle formation and structural interlocking at the interface. Owing to the hexagonal pattern formed by the graphene units, the surface energy landscape showed higher variation in the armchair direction. This forces the adjacent polymers to bypass the high-energy points on the potential energy surface, thus elongating the displacement trajectories. By contrast, the zigzag orientation maintains good interfacial bonding with the polymer under an external load owing to its high flexibility and low surface-energy variation. These findings provide molecular-level insights into the chirality-induced anisotropy in vibration damping and highlight a novel design strategy for the optimization of the dynamic mechanical performance of graphene-reinforced nanocomposites.

1. Introduction

The incorporation of nanoscale reinforcements into polymers has emerged as a highly effective strategy for improving the poor intrinsic mechanical performance of polymers over a long period of time [1–4]. In particular, notable progress has been achieved in experimental efforts to improve the energy dissipation and vibration-damping characteristics of polymers through the creation of nanocomposite systems. Vibration damping is known to be mainly governed by energy dissipation at the interface between the reinforcement and matrix. Friction loss and slippage occur at the interface owing to the inherent stiffness differences between the components of heteromaterials, imperfect bonding, and stress propagation discontinuities. Nanocomposite systems for vibration damping exploit imperfect interface states to facilitate energy dissipation, especially under dynamic loading conditions. In particular, as the reinforcement approaches the nanoscale, the particle-matrix contact area changes significantly with subtle changes in the size of the reinforcement particles [5]; thus, mechanics at the interface play a critical role in determining the properties of the entire macrosystem. Such particle-size-based interface design methods can achieve improved properties without significantly increasing the overall material density. Therefore, nanocomposites are of particular interest in advanced engineering applications, such as aerospace and automotive systems, where both lightweight structures and effective vibration damping are required [6–8].

Graphene is one of the most promising reinforcement materials owing to its unique two-dimensional geometry that allows for maximum interfacial area, particularly for applications related to energy

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dissipation [9–12]. The atomically thin, hexagonally arranged carbon lattice of graphene allows for a large filler-matrix contact area even at small loadings. In fact, epoxy systems incorporating graphene exhibit significantly greater improvements in the damping ratio (26 %–128.9 % at 0.025–1.2 wt %) [13,14] compared to those reinforced with silica (15 %–42 % at 1.0–3.5 wt %) [15,16], carbon fiber (12.2 %–39.51 % at 1.13–5.0 wt %) [17], and carbon nanotubes (41.1 %–82 % at 0.4–0.5 wt %) [18,19]. The superior energy dissipation of polymer-graphene nanocomposites is due to the stick-slip behavior of the interface [20]. Under cyclic mechanical loading exceeding the critical limit, the repetitive destruction and reformation of the mechanical interlocking and van der Waals forces at the interface have been shown to significantly improve the damping properties of the entire system. In addition to stick-slip behavior, interfacial debonding and slippage are also recognized as key damping mechanisms in nanocomposites. Multiscale studies have shown that graphene orientation and functionalization govern debonding-induced slip, which effects energy dissipation and structural failure under cyclic loading [21,22]. Moreover, previous studies have consistently highlighted the critical role of interfacial interactions in governing the performance of nanocomposites [23,24]. Factors such as covalent bonding [25,26], filler geometry [27–29], and environmental conditions [30] have been shown to strongly influence load transfer, interphase stiffness, and stress dissipation. For instance, functionalized or grafted fillers enhance interfacial bonding and significantly improve mechanical properties, while variations in filler morphology and alignment modulate interphase thickness and efficiency of stress transfer [31,32].

The development of strategies that maximize the intrinsic properties of the interface is the most important current challenge for the optimal design of graphene nanocomposites. Numerous studies have demonstrated a mismatch between the intrinsic mechanical properties of graphene and the resulting composite performance. For instance, even though armchair graphene exhibits higher intrinsic stiffness, nanocomposites reinforced with zigzag graphene often exhibit superior tensile behavior [33–37]. Such discrepancies highlight the fact that interfacial phenomena, which are strongly influenced by the chirality of graphene, play a more dominant role in the overall mechanical behavior of the composite than the intrinsic properties of the filler. In other words, the relative directionality between the direction in which the external vibrational load was applied to the composite and the chirality of the inserted graphene significantly influences the mechanical properties of graphene nanocomposites. Nevertheless, the relationship between graphene chirality and energy dissipation and its underlying mechanisms has not been sufficiently studied. Nanocomposite design that exploits this relationship has become particularly important in recent years because control of the alignment, stacking configuration, and chirality of graphene during composite fabrication has become feasible [38,39].

This study investigated the influence of the chirality-specific alignment of graphene on the nanoscale energy dissipation mechanisms in polymer nanocomposites using all-atom molecular dynamics (MD) simulations. The correlation between the relative directionality of the applied external vibration load and the inserted graphene and the vibration damping properties of the entire system were systematically analyzed at the atomic scale. The anisotropic properties of the polymer domains in the interfacial region influenced by the chirality of graphene were quantitatively analyzed. In particular, the structural interlocking of the graphene-polymer interface and slippage of polymer chains during the shear loading process have been explicitly elucidated. This study provides directions for future design of graphene nanocomposites that maximize vibrational damping, as well as deep physicochemical insights into the damping mechanism at the interface.

2. Models and methods

2.1. Cross-linked epoxy and epoxy/graphene microstructure modeling

The epoxy microstructure was modeled in an all-atom MD simulation environment. The diglycidyl ether of bisphenol A (DGEBA) and diethyl toluene diamine (DETDA) were used as the epoxy resin and hardener, respectively. Initially, the resin and hardener molecules were randomly placed in a periodic cubic cell at a molecular ratio of 2:1 (300 and 150 molecules, respectively). To avoid unintended interference from interactions between the molecules during the initial arrangement, the unit cell was prepared at a very low density (0.1 g/cm^3). The unit cell dimensions were then uniformly reduced in all directions to gradually increase the density to 1.0 g/cm^3. All covalent and non-covalent interactions of the molecules inserted within the unit cell were described using the polymer-consistent force field (PCFF) [40]. Energy minimization was carried out using the conjugate gradient method with an energy cutoff of 0.001 kcal/mol for every 0.1 g/cm^3 increase in the density. Once the system reached the target density, the NVT (1 ns, 300 K) and NPT (1 ns, 1 atm, 300 K) ensembles were conducted sequentially to further stabilize the model. Epoxy/graphene nanocomposite models were prepared using the same procedure as that used for epoxy simulations, differing only in that during the initial molecular positioning, a graphene monolayer with a size of 5.2 × 5.1 nm^2 and a filler content of 8.63 wt % was initially placed at the center of the unit cell. To eliminate dangling bonds at the edges, all edge carbon atoms of the finite graphene sheets were terminated by hydrogen atoms, which stabilizes the structure by saturating the dangling bonds and completing the sp^2 orbital configuration. The graphene nanofillers were aligned parallel to the xy plane without folding or bending. The chiral graphene was arranged such that the direction of the armchair was parallel to the x-axis of the entire unit cell. Accordingly, the direction in which the chirality forms a zigzag corresponds to the y-axis of the unit cell.

Crosslinking between the resin and hardener molecules was performed to generate cross-linked networks. During the simulation, the locations and relative distances of all reactive sites were continuously monitored. When the distance between the reactive sites was within the cutoff value, a new covalent bond was formed between them, and the associated force field parameters were updated. The initial cutoff distance was set to 3 Å and was incrementally increased by 0.5 Å until the network reached a final crosslinking ratio of 75 %. Energy minimization was performed for the entire system whenever crosslinking occurred. After reaching the desired level of crosslinking, the system was sequentially subjected to thermodynamic equilibration in the NVT ensemble (2 ns, 300 K) and in the NPT ensemble (2 ns, 1 atm, 300 K). Subsequently, the model was annealed at a high temperature of 550 K for 2 ns and gradually cooled to 300 K over 4 ns. This process was intended to eliminate any structural distortions that could occur during the crosslinking process. The completed structure of the epoxy/graphene model is shown in Fig. 1**(a)**.

The other simulation details are as follows. The commercial MD software Materials Studio 2020 (Dassault Systèmes, France) was used to construct amorphous neat epoxy and epoxy/graphene nanocomposite models. Static and dynamic mechanical tests were conducted using the open-source MD code LAMMPS (Sandia Labs, USA).

2.2. Static mechanical properties of neat epoxy and epoxy/graphene nanocomposites

To examine the accuracy of the predictions of the mechanical behavior of the constructed model, the relevant material properties were first evaluated through static mechanical tests. While the unit cells were deformed at an engineering strain rate of 10^9/s, the atomic stress tensor (σ) of the microstructures was calculated based on the virial theorem, which can be expressed as follows:

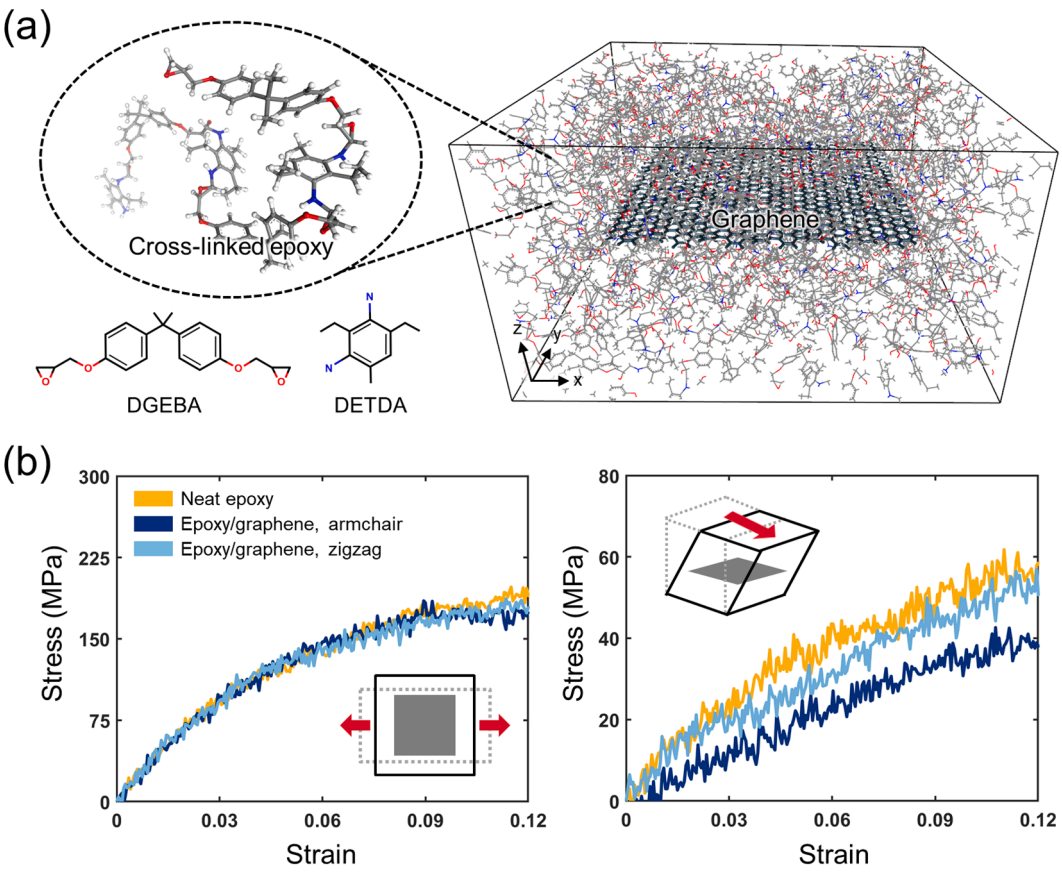


Fig. 1. Molecular-level modeling of neat epoxy and epoxy/graphene nanocomposites. (a) All-atom configuration of cross-linked epoxy integrated with a single graphene sheet. (b) Static mechanical tests of the cross-linked epoxy models. The external forces were applied in the in-plane tensile direction (left) and out-of-plane shear direction (right), respectively.

$$\boldsymbol{\sigma} = \frac{1}{V}\left(-\sum_{i}^{N} m_i(\boldsymbol{v}_i \boldsymbol{v}_i^T) + \frac{1}{2}\sum_{i}^{N}\sum_{j\neq i}^{N} \boldsymbol{r}_{ij}\boldsymbol{F}_{ij} \right) \quad (1)$$

where V is the volume, N is the number of constituent atoms, m_i and $\boldsymbol{v}_i$ are the mass and velocity of the i th atom, respectively, and $\boldsymbol{r}_{ij}$ and $\boldsymbol{F}_{ij}$ indicate the relative position and interaction force between the i th and j-th atoms, respectively [41]. It should be noted that the strain rate used in the simulations (10^9 /s) is several orders of magnitude higher than typical experimental strain rates. This limitation is inherent to MD simulations due to nanosecond time scales, and while it may affect the absolute values of mechanical properties, it remains a standard practice in MD studies. Molecular dynamics simulations inherently operate at nanosecond timescales, requiring high strain rates that can restrict full polymer chain relaxation and potentially underestimate viscoelastic behavior. To mitigate such artifacts, we adopted two technical strategies. First, instead of applying strain only through box deformation, the coordinates of all atoms were uniformly rescaled, ensuring homogeneous deformation across the epoxy network. This approach, combined with the high crosslinking ratio of our epoxy model, allows sufficient molecular rearrangement even within short timescales. Second, to preserve physical consistency under high strain rates, we employed the polymer consistent force field (no bond breakage permitted) together with Nosé-Hoover thermostat/barostat control, maintaining constant temperature and pressure. These measurement techniques allow materials to exhibit a quasi-static-like behavior even in nanosecond timescale environments. We also show that the considered strain rate is sufficient to produce a converged quasi-static response of the considered microstructure (see **Fig. S1**).

The mechanical deformation was applied along the in-plane tensile and out-of-plane shear directions, where the mechanical effects of graphene were most prominent due to the maximized projected area [42]. In particular, to examine the influence of graphene chirality, the results obtained by the nanocomposites model for the armchair and the zigzag directions were analyzed.

The stress–strain curves of the neat epoxy and epoxy/graphene nanocomposites are shown in Fig. 1**(b)**. The elastic and shear moduli were calculated from the slopes of the stress–strain curves for regions where the strain was <3 % and are summarized in Table 1. The simulation results indicated that the mechanical properties of the neat epoxy model were in close agreement with those of previous studies on materials with the same composition [13,43,44]. Therefore, we conclude that the force field and equilibrium method applied in the current MD simulations adequately describe the mechanical response of the material. Meanwhile, for the nanocomposite model, the tensile modulus showed no significant difference from that of the neat epoxy. The lack of improvement in the tensile modulus is due to incomplete interfacial bonding between graphene and the epoxy matrix, which has been reported in many previous studies [45]. Predominantly sp^2-bonded nanocarbon materials form slippery interfaces with adjacent polymer chains, resulting in inefficient stress transfer and limited reinforcement properties [46]. Moreover, graphene contributes little reinforcement because only a narrow portion of its surface engages with the stress, and weak interfacial adhesion promotes slip rather than efficient load transfer. However, there is a difference where the tensile moduli is slightly higher for the zigzag orientation compared to the armchair orientation, which is consistent with previous studies [35–37]. Interestingly, compared to that of the neat epoxy, a significantly reduced stress profile was observed for the nanocomposite model in the out-of-plane shear tests, and the anisotropy along the shear direction

Table 1
Structural and mechanical properties of the considered epoxy models.

	Approach	Resin/hardener	Cross-linking ratio (%)	Density (g/cm^3)	Elastic modulus (GPa)	Shear modulus (MPa)
Epoxy	MD simulation	DGEBA/DETDA (300/150)	76.8	1.095	3.02	807.34
Epoxy/graphene, armchair			76.2	1.12	3.04	370.67
Epoxy/graphene, zigzag					3.12	655.58
Epoxy [13]	Experimental	DGEBA/DETDA	–	1.08	2.73	–
Epoxy [43]	MD simulation		80	1.08	2.75	991.84
Epoxy [44]			80.7	1.107	3.73	–

was also significant. Specifically, the degradation of the mechanical response was more pronounced when shear strain was applied in the armchair direction of graphene. These findings highlight the significant influence of graphene chirality on the mechanical behavior, such as energy dissipation, motivating further investigation of the deformation behavior. We have additionally verified the shear modulus obtained from the simulations by comparing them to experimental results (see **Table S1** and **Table S2** from **Supplementary Note 1**).

3. Results and discussion

3.1. Oscillatory shear simulation of epoxy/graphene nanocomposites

Oscillatory shear simulations were performed to investigate the dependence of the viscoelastic behavior of graphene on its chirality (Fig. 2**(a)**). The polymer consistent force field (PCFF) was employed to represent both bonded and non-bonded interactions among the molecules in the unit cell. Non-covalent attractions and repulsions between atoms are determined based on the sixth-power combination rule, as defined by the PCFF. Also, the non-bonded interactions were modeled using a Lennard-Jones (9–6) and Coulombic cutoff of 10 Å, with long-range electrostatics treated using the Ewald summation method. In the ensemble, a Nose-Hoover thermostat and barostat are applied to preserve internal equilibrium. The equilibrated unit cell at 300 K was quenched to 0.1 K for 500 ps before shear deformation, ensuring that virial stresses during subsequent mechanical testing originated from interatomic potential energy rather than thermostat artifacts. During oscillatory shear, shear amplitudes and periods were set to produce maximum shear strains and the shear frequency. The model was subjected to a sinusoidal shear strain, as follows:

$$\gamma(t) = \gamma_0 \sin(\omega t) \tag{2}$$

where γ_0 and ω are the maximum shear strain and oscillation frequency of the damping test, respectively [47]. The maximum shear strain values ranging from 2 % to 10 % were applied to the unit cell in the NVT ensemble. The shear frequency was set to 2×10^{10} Hz, similar to that used in the experimental quartz crystal microbalance measurements [48]. Although this frequency is significantly higher than that typically used in conventional dynamic mechanical analysis, such values are commonly adopted in molecular dynamics (MD) simulations due to intrinsic limitations in accessible timescales [49–52]. Furthermore, simulation conditions including affine remapping of atomic coordinates to ensure homogeneous deformation and strict thermostat/barostat control to maintain the target thermodynamic state were implemented to eliminate artifacts associated with high strain rates. Therefore, the loss factors obtained in this study are in good agreement with experimental results reported for epoxy/graphene oxide and epoxy/graphene nanoplatelet composites, despite differences in absolute frequency [14, 53,54]. This consistency supports the reliability of the chosen frequency and confirms that the relative damping trends captured in our simulations are governed by intrinsic interfacial and structural mechanisms rather than the specific frequency applied.

The stress response of the system during the simulation was derived based on the virial theorem and was fitted to a sinusoidal function:

$$\tau(t) = \tau_0 \sin(\omega t + \delta) \tag{3}$$

where τ_0 is the maximum amplitude of the stress and δ is the phase lag between the stress and strain profiles. δ is a material-specific value derived from the viscoelastic properties of the epoxy matrix, and its

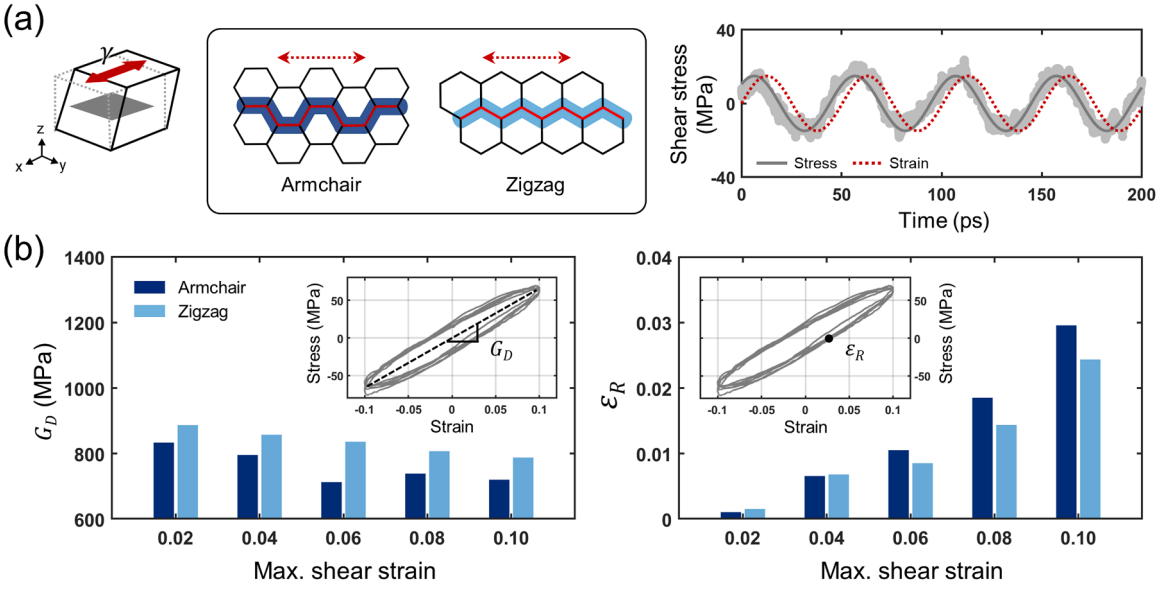


Fig. 2. Energy dissipation characteristics of epoxy/graphene nanocomposites for different loading directions. (a) Schematic of oscillatory shear simulations along armchair and zigzag directions (left) and the associated stress–strain response (right). (b) Comparison of G_D and ε_R for deformations along armchair and zigzag directions. Insets show stress–strain hysteresis loops under cyclic loading.

calculation process through MD simulations and physical meaning have been thoroughly discussed in previous studies [30].

The simulation model was subjected to six cycles of sinusoidal shear strain, and the averaged stress response was used to construct stable hysteresis loops. The load transfer and energy dissipation performance of the nanocomposites were evaluated using two independent parameters: the dynamic shear modulus (G_D) and the residual strain (ε_R). G_D was determined from the slope of the major axis of the hysteresis loop that characterizes the stress–strain response:

$$G_D = \frac{\Delta\tau}{\Delta\gamma} = \frac{\tau_{\max} - \tau_{min}}{\gamma_{\max} - \gamma_{min}} \tag{4}$$

where τ and γ represent shear stress and strain, respectively. Meanwhile, ε_R is determined from the molecular rearrangement and inelastic properties of the system when the applied load is removed. For all cases considered in this study, the stress–strain curves under cyclic loading exhibit hysteresis loops due to the viscoelastic properties of the model. Accordingly, ε_R can be evaluated as the strain value of the hysteresis loop at the applied stress of zero.

When cyclic loading was applied along the out-of-plane axis of graphene, a clear dependence on graphene chirality was observed for both G_D and ε_R, as shown in Fig. 2**(b)**. In particular, we found that the mechanical response showed completely different behavior with the changes in the relationship between the load direction and graphene chirality. Across all maximum shear strains, the zigzag direction consistently exhibited a higher dynamic shear modulus, indicating its superior load-transfer efficiency. By contrast, the armchair direction displayed a larger residual strain, suggesting greater energy dissipation capabilities. This indicates that the energy-dissipation properties of the interfacial region of the nanocomposite varied significantly with the changes in the direction of the inserted graphene and applied shear load. Such anisotropic properties demonstrate that graphene orientation plays a key role in the design of polymer nanocomposites for vibrational damping. In Fig. 3, a comparable trend is observed for the phase lag and loss tangent, consistent with the results discussed above. Moreover, the predicted loss factor range from our simulations aligns quantitatively with experimental data (**Table S3**), providing additional validation of the methodology.

3.2. Analysis of spatially different mechanical behavior of epoxy matrix inside nanocomposites

In the previous sections, we obtained important results regarding the chirality-dependent mechanical properties of nanocomposites. To further elucidate this behavior, we performed a more detailed analysis of the mechanical response in localized regions within the nanocomposite model. The polymer matrix interacting with the heterogeneous nanofiller forms a strongly adsorbed layer near the interface (i.e., the interphase), which alters its structural and mechanical properties [55]. Therefore, the area occupied by the epoxy matrix in the nanocomposite model was virtually separated into regions of the interphase and the bulk domain. The boundary between the interphase and the bulk matrix was defined based on the density distribution of the epoxy (see **Fig. S2** of **Supplementary Note 2**).

To characterize the spatial mechanical behavior, the atomic strain tensor of the epoxy matrix was obtained during the oscillation cycle based on the finite strain theory. The Green-Lagrange strain tensor ($\boldsymbol{E}$) evaluates the relative displacement between atoms while excluding translational or rotational motion [56]. Thus, the evaluation of $\boldsymbol{E}$ enables clear visualization of the discontinuous strain field that appears at the domain boundary during deformation. The Green-Lagrange strain tensor is calculated as follows [57]:

$$\boldsymbol{E} = \frac{1}{2}\left(\boldsymbol{F}^T\boldsymbol{F} - \boldsymbol{I}\right) \tag{5}$$

where $\boldsymbol{F}$ is the deformation gradient tensor, and $\boldsymbol{I}$ is the identity tensor. The deformation behavior of the interphase and bulk region was evaluated using the von Mises strain (ε_{vM}), which is defined as:

$$\varepsilon_{vM} = \frac{\sqrt{2}}{3}\sqrt{(\varepsilon_1 - \varepsilon_2)^2 + (\varepsilon_2 - \varepsilon_3)^2 + (\varepsilon_3 - \varepsilon_1)^2} \tag{6}$$

where ε_1, ε_2 and ε_3 are the principal strains.

The Green-Lagrange strain tensor and von Mises strain originate from continuum mechanics. Although the microstructural models currently considered lie in the nanoscale region, the approach can be consistently applied because the time-dependent displacements and local strain gradients of all constituent atoms are fully defined in the simulation. Specifically, we determined the strain tensor by deriving the affine transformation of each atom for its neighboring atoms. The local strain was computed by considering atoms within a cutoff radius (15 Å) around each reference atom as vertices of a three-dimensional polyhedral domain. The initial and deformed configurations of this local domain were compared through an affine transformation, so that changes in atomic positions under loading were interpreted as shape changes of the polyhedron. Such a geometric approach allows for consistent quantitative mapping of the displacements of individual atoms while maintaining nanoscale resolution.

The von Mises strains of the interphase and bulk were obtained individually during the application of a single oscillation cycle to the unit cell (Fig. 4**(a)**). The results indicated that the von Mises strain of the interphase was higher than that of the bulk region under both armchair and zigzag loading conditions. This suggests that the energy-dissipation enhancement of the nanocomposites originates predominantly from graphene-epoxy slippage. Specifically, the interphase matrix exhibits a higher atomic strain under armchair-direction loading than under zigzag loading. In other words, the anisotropic energy dissipation and load transfer behaviors of the nanocomposites were due to chirality-dependent interfacial features. This is further supported by the von Mises strain distribution (Fig. 4**(b)**). While the strain in the bulk region was the same regardless of the loading direction, the interfacial matrix component exhibited a higher displacement against the graphene filler during loading in the armchair direction.

We also analyzed the von Mises strain over multiple loading cycles (Fig. 4**(c)**). The results indicate a greater von Mises strain accumulation

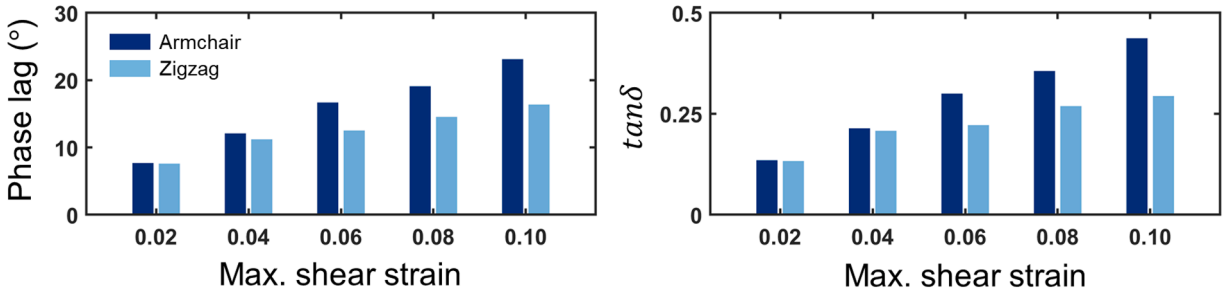


Fig. 3. Quantification of the phase lag (left) and damping coefficient (right) in different loading directions of epoxy/graphene nanocomposites.

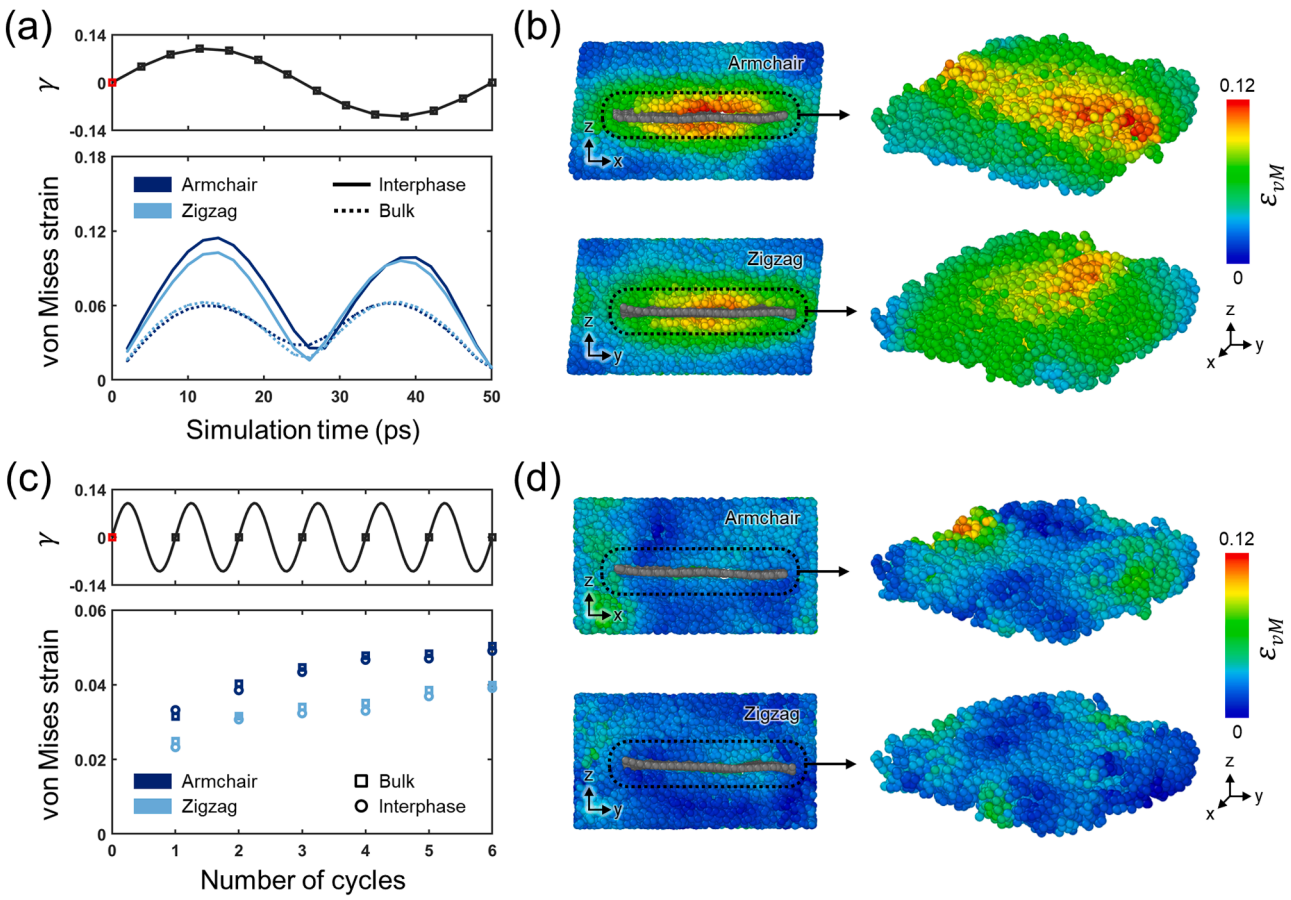


Fig. 4. Strain field analysis of the interfacial region around graphene. (a) Changes in the atomic strain distribution during a single oscillation cycle. (b) Visualization of the atomic strains under deformations in the armchair (top) and zigzag (bottom) directions. Snapshots were taken at the maximum shear strain at 12.5 ps. (c) Changes in the atomic strain distribution over multiple oscillation cycles. (d) A representation of the atomic strains under deformations in the armchair (top) and the zigzag (bottom) directions. Snapshots were taken at the final frame of the simulation.

in the armchair direction than in the zigzag direction, which is consistent with previous observations. This further reinforces the conclusion that the armchair direction exhibits lower load transfer efficiency and higher energy dissipation. In particular, we found that at the end of each cycle, the residual strain was more pronounced in the bulk region than in the interphase. The graphene nanofiller adsorbs the epoxy components onto its surface, leading to a locally high cross-linking density in its vicinity [58]. As a result, the interface exhibited a locally enhanced structural resilience compared to that of the bulk region (Fig. 4**(d)**).

3.3. *Chirality-dependent energy dissipation in nanocomposites*

The results described above demonstrate that graphene-functionalized epoxy nanocomposites exhibit significant chirality-dependent differences in energy dissipation and load transfer. This behavior is primarily attributed to the anisotropic interactions at the heterointerface. In this section, we identify two molecular origins of these anisotropies, namely wrinkles and surface patterns of graphene.

Generally, the wrinkling of graphene significantly affects the mechanical behavior of nanocomposites. The discontinuous interface formed by wrinkled graphene enhances the load transfer efficiency and

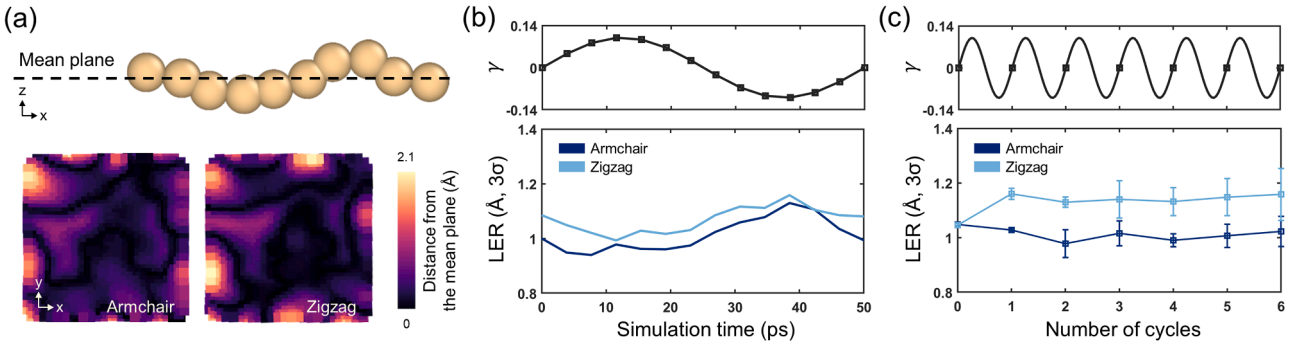


Fig. 5. Influence of graphene chirality on wrinkles under shear oscillation tests. (a) Schematic of the calculation of the mean plane deviation (top) and the visualization of atomic displacement from the mean plane (bottom). (b) Variation of applied shear strain and line edge roughness (LER) of graphene during single and (c) multiple oscillation cycles. Error bars represent the standard deviation of the properties obtained from three independently prepared models.

prevents interfacial slippage [59,60]. Within the polymer nanocomposites, a structural interlocking effect is formed as adjacent matrix components penetrate wrinkles and physically bond to each other. To quantify the wrinkling, we calculated the displacement deviation of each constituent atom from the mean plane of graphene (Fig. 5**(a)**). The atomic displacement visualization at the end of the simulation revealed brighter highlights in the zigzag direction, indicating a greater degree of wrinkling. Such anisotropic properties were also evident from the roughness values of the graphene surface. For the inserted graphene, the line-edge roughness (*LER*) was quantified using the following equation:

$$LER = 3\sqrt{\frac{1}{N}\sum_{i=1}^{N}(l_i - \bar{l})^2} \tag{7}$$

where N is the number of atoms of the graphene, and l is the relative distance between each atom and the mean plane. The analysis was performed for all sections during the damping test, thus allowing quantitative tracking of the evolution of wrinkles under cyclic loading.

When a single oscillation is applied to the nanocomposite system, the LER profile exhibits a sinusoidal shape (Fig. 5**(b)**). In other words, the initial shear deformation flattened the graphene sheet, and the subsequent recovery to the undeformed state resulted in the roughening of the wrinkles. Interestingly, the magnitude of the wrinkles under armchair deformation was always smaller than that in the zigzag direction. This suggests that the interlocking effect that affects the matrix adjacent to the surface of graphene is less pronounced in the armchair direction. The flatter conformation of graphene preserves its initial structure under loading, making intermolecular slippage more effective. This feature is preserved even after multiple cycles (Fig. 5**(c)**). Note that the orientation-dependent wrinkling characteristics of graphene are not a consequence of the limited size or initial conditions of the model, but rather are a consequence of the inherent differences in the stiffness of the system. Three epoxy/graphene models were prepared and repeated with the same oscillatory shear test, which always resulted in a flatter surface in the armchair direction, facilitating energy dissipation.

Wrinkles and surface properties of graphene can be further tuned through intentional manipulation of vacancies, adatoms, Stone-Wales defects, and dislocations, which in turn govern energy dissipation. Future work will focus on establishing the correlation between wrinkling at the interface and energy dissipation by simultaneously considering the structural defects and anisotropic properties of graphene itself.

The differences in wrinkle configurations were further examined by comparing the intrinsic stiffness of graphene. Infinite graphene models periodically connected in both x and y directions were prepared and equilibrated using simulations under the NPT ensemble at 300 K and 1 atm carried out for 1 ns. Subsequently, a uniaxial tensile strain was applied in each direction to estimate the mechanical anisotropy of the microstructure. The results demonstrated that graphene has higher mechanical stiffness in the armchair direction (Fig. 6**(a)**), with Young's modulus values of 1.01 TPa and 0.98 TPa obtained in the armchair and zigzag directions, respectively. The higher modulus in the armchair direction found in our study is consistent with the results obtained in the previous studies on pristine graphene sheets [61,62]. The increased structural rigidity suppresses the out-of-plane vibrations of the constituent atoms, leading to a flatter wrinkle profile. For further verification, we estimated the vibrational characteristics of graphene (Fig. 6**(b)**). The relative displacements of graphene atoms were tracked during equilibration in the NVT ensemble at the intervals of 0.3 ps. The scatter plots illustrate the atomic displacement distributions, where an elliptical fit was used to quantify the fluctuation range with a 95 % confidence interval. In the armchair mode, the x-axis represents the displacement along the armchair direction, whereas the y-axis represents the out-of-plane displacement. Similarly, for the zigzag mode, the x-axis corresponds to the displacement in the zigzag direction, and the y-axis corresponds to the out-of-plane displacement. It is clear that the out-of-plane vibration is stronger in the zigzag direction with weak stiffness (i.e., higher 'b' value in Fig. 6**(b)**). This shows that the high intrinsic stiffness maintains a flatter sheet by suppressing vertical vibrations at a given thermodynamic energy.

The surface pattern of graphene is another origin of anisotropic energy dissipation. Polymer segments interacting with the graphene sheet exhibit strong adsorption at the center of the hexagonal structure [63]. Interfacial energy dissipation occurs when adsorption is disrupted by an external force. For a detailed analysis of chirality-dependent absorption, we obtained the van der Waals energy profile on the graphene surface. A single carbon atom drifting on the surface of a graphene sheet was prepared, and its interfacial interaction energy was continuously monitored. The resulting energy landscape shows a clear chirality dependence of the energy landscape (Fig. 7**(a)**). The lowest energy was found at the center of the hexagonal lattice, and the energy increased with the proximity of graphene atoms and carbon. Importantly, atoms located above the lattice center must traverse a very steep energy gradient to move along the armchair pattern, suggesting higher energy dissociation (profile (1) in Fig. 7**(a)**). By contrast, the zigzag pattern allows adjacent atoms to pass through the energy valleys formed by the graphene atoms, allowing them to maintain a relatively stable interfacial adsorption state (profile (2) in Fig. 7**(a)**).

The differences in the energy profiles significantly affect the displacement characteristics of the polymer chains in the interfacial region under mechanical deformation. The displacement distribution of the interfacial matrix atoms was obtained when sinusoidal shear strain was applied to the nanocomposite unit cell (Fig. 7**(b)**). The polymer components exhibited linear atomic trajectories in the load-parallel direction under zigzag deformation. The energy valleys allowed the polymer components to slide while preserving stable interfacial bonds. By contrast, molecules undergoing deformation along the armchair

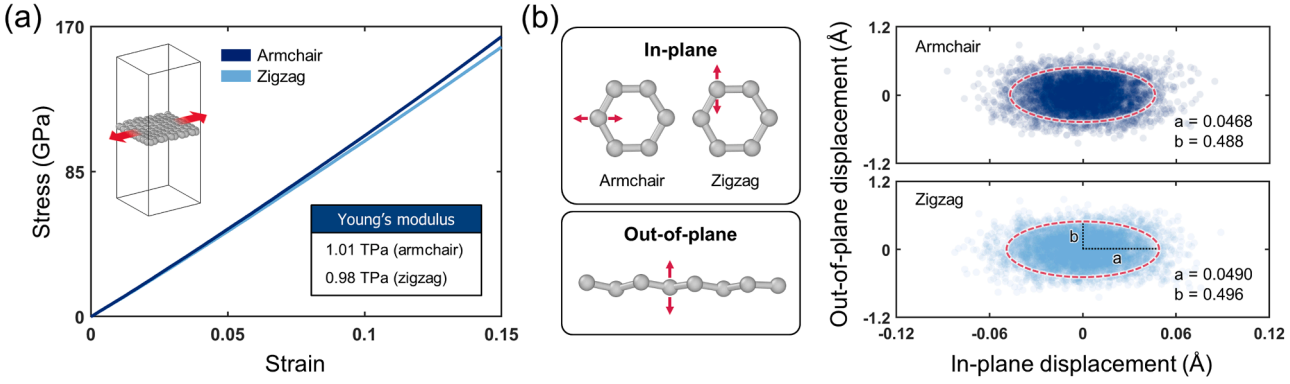


Fig. 6. Chirality-dependent mechanical stiffness of graphene. (a) Stress–strain profiles of infinite-length graphene under tensile loading. (b) Schematics of in-plane and out-of-plane displacement calculations (left) and vibration mode analysis of graphene atoms (right).

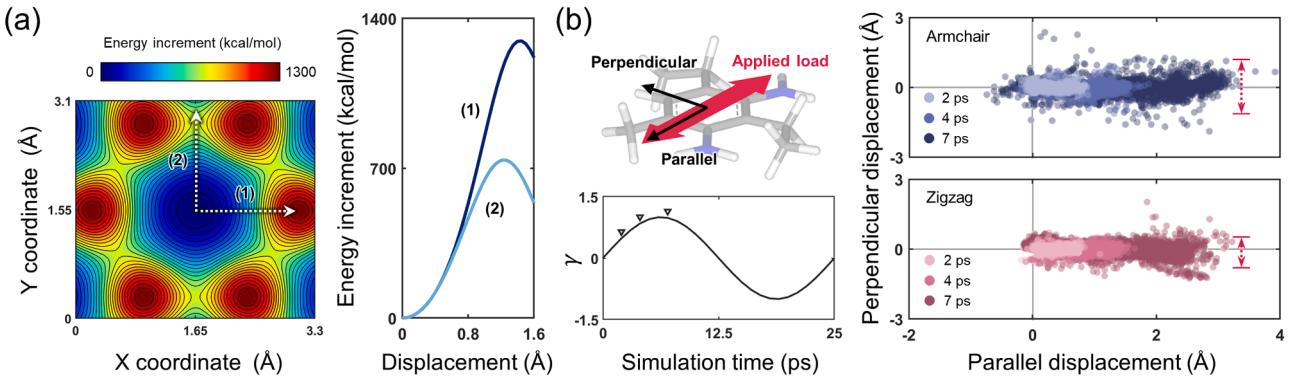


Fig. 7. Influence of graphene structure on the behavior of adjacent polymer chains. (a) Energy landscape of the graphene surface (left) and its profile along the sliding pathway (right). The results of the comparison of the profiles under expansion of the graphene area are presented in **Fig. S3**. (b) Directional displacement of the epoxy components forming the interface under a process in which the entire system experiences a sinusoidal shear strain. The direction of displacement experienced by the polymer component is defined relative to the loading direction (top left). The points at which displacements were evaluated during the sinusoidal strain are highlighted (bottom left). The accumulated displacements at these points are plotted (right).

direction deviate significantly from the linear trajectory and avoid high-energy positions. In other words, the absolute magnitude of the displacement of the polymers in the interfacial region was amplified as they fluctuated diagonally rather than parallel to the load. In summary, the polymers sheared along the armchair pattern direction exhibited longer molecular trajectory motions than those sheared along the zigzag pattern. This explains why the nanocomposites exhibited relatively high energy dissipation under shear deformation in the armchair pattern direction.

To strengthen the conclusion regarding the role of interfacial effects in damping improvement, we additionally performed the pull-out simulations. One side of the epoxy/graphene model was opened along the desired pull-out direction, while the epoxy matrix was fully constrained (Fig. 8**(a)**). The pull-out tests were performed by applying a constant velocity of 10^{-4} Å/fs to one end of the graphene sheet in an NVT ensemble at 0.1 K. During the separation process, the variation in the interfacial energy (E_{int}) was continuously measured using the equation:

$$E_{int} = E_{total} - E_{epoxy} - E_{graphene} \tag{8}$$

where E represents the potential energy and the subscripts denote the respective system components. Subsequently, the pull-out force (F) was computed as:

$$F = \frac{\Delta E_{int}}{\Delta d} \tag{9}$$

where ΔE_{int} is the change in interfacial energy and Δd incremental displacement of graphene. The atomic forces evolving on graphene during pullout simulations consist of three distinguishable stages (Fig. 8 **(b)**) [64]. Stage I: The pull-out force exhibits a steep and rapid increase corresponding to the onset of interfacial resistance. Due to the wrinkle configuration, the magnitude of the peak force in this stage is slightly higher than that which converges thereafter. Stage II: After reaching the peak point, the pull-out force converges steadily. The profile is not stable and fluctuates significantly due to the continuous destruction and reformation of the interfacial bonds. Stage III: The pull-out force profile drops to zero, implying that the distance between graphene and epoxy has exceeded the cut-off radius for van der Waals interactions.

Given that energy dissipation in polymer nanocomposites is predominantly governed by interfacial sliding mechanisms, the focusing primarily on Stage I reveals the characteristics of the initial detachment process. In other words, the pull-out force at Stage I corresponds to the energy required to overcome the static friction between graphene and epoxy. Importantly, the peak value was lower in the armchair direction, indicating a lower energy barrier for interfacial debonding. These observations confirm that the interfacial debonding and sliding mechanisms revealed by the pull-out simulations support our argument that interfacial effects contributes to the observed damping improvements.

3.4. Comparative analysis of monolayer and bilayer graphene fillers

In addition to the single-layer graphene filler system, we also performed simulations of epoxy nanocomposites reinforced with a bilayer graphene filler to further validate the role of graphene morphology in damping performance. The bilayer filler was modeled in an AB-stacked configuration, following the same procedure as described in Section 2.1 (Fig. 9**(a)**). The density of the bilayer model is 1.16 g/cm^3, which is higher than the density of the neat epoxy and epoxy/graphene monolayer model (1.095 and 1.12 g/cm^3 respectively). Such an increase is

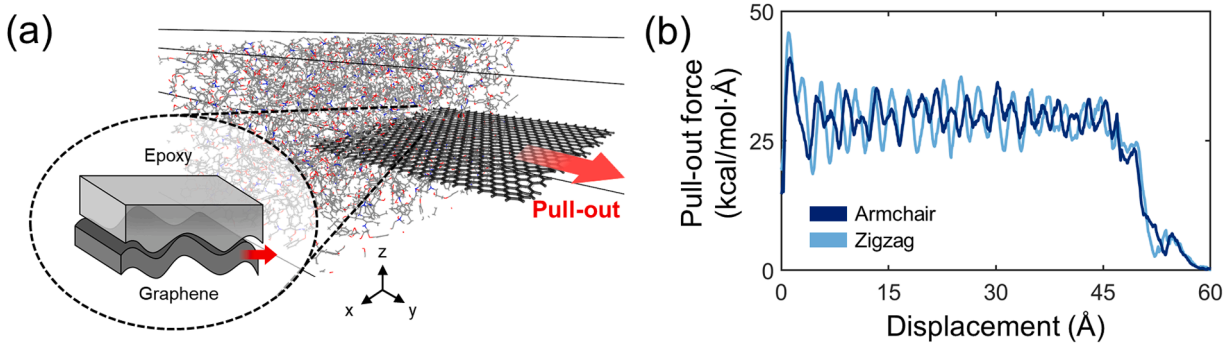


Fig. 8. Pull-out simulation to observe the interfacial properties between the graphene and epoxy. (a) Schematic representation of the pull-out test. (b) Pull-out force profiles for graphene in the armchair and zigzag directions.

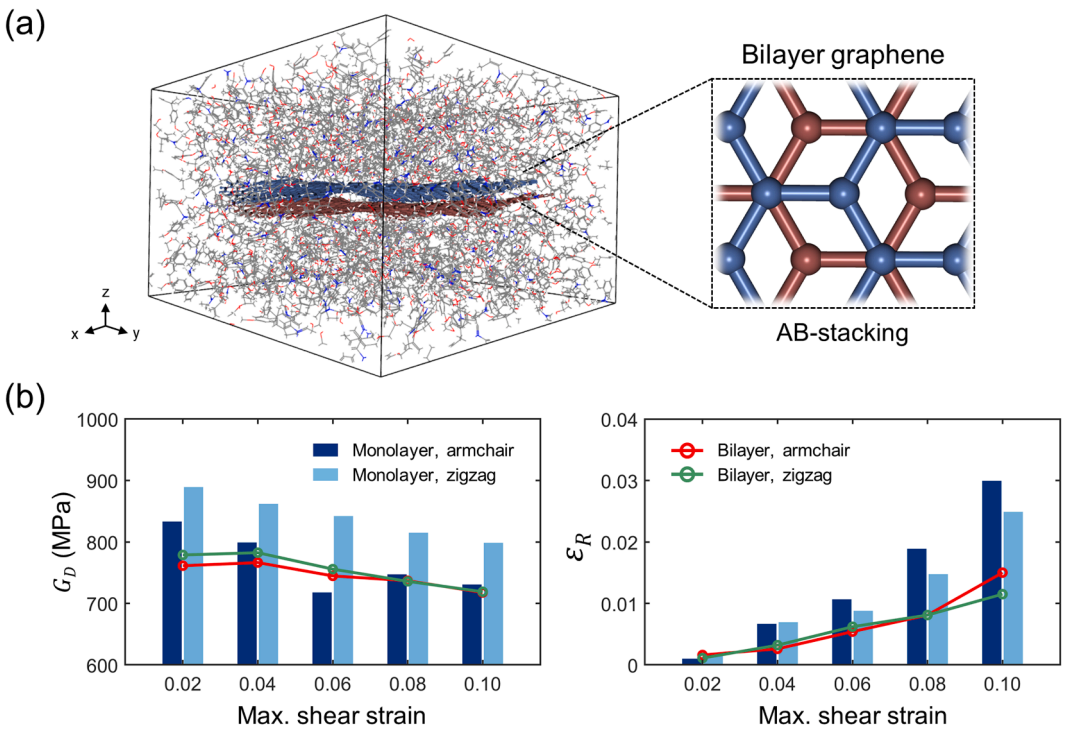


Fig. 9. Damping responses in monolayer and bilayer graphene. (a) Schematic of a nanocomposite model with bilayer graphene sheets embedded in the epoxy matrix. (b) Comparison of G_D and ε_R for monolayer and bilayer graphene.

attributed to the higher filler content and the more compact stacking of graphene layers. The damping performance was then evaluated under oscillatory shear loading, and the results are summarized in Fig. 9**(b)**. Two major differences emerged when compared to the monolayer system. First, the orientation dependence, which was a central finding in the monolayer case, became significantly negligible in the bilayer case. This is attributed to the increased bending rigidity of the AB-stacked double layers, which suppresses the formation of wrinkles in all directions. Second, the overall magnitudes of both G_D and ε_R were reduced relative to those of the monolayer composites. This reduction originates from two factors: (1) the diminished mechanical interlocking between the epoxy matrix and graphene surface due to the smoother, flatter morphology of bilayer graphene, which weakens stress transfer across the interface, and (2) the strongly adsorbed interfacial region stabilized by enhanced van der Waals interactions, which exhibits a more elastic response and resists structural rearrangements under cyclic loading. By contrast, the monolayer graphene, with its higher surface-to-volume ratio and more adaptable interfacial area, provides stronger interfacial coupling and enables more efficient stress transfer. These results collectively suggest that while bilayer graphene enhances the structural compactness of the composite, it is the monolayer graphene that offers superior vibrational energy dissipation, primarily due to its ability to accommodate interfacial deformation and facilitate energy dissipation through wrinkling.

Recent studies have demonstrated that the alignment, orientation, and chirality of graphene in polymer nanocomposites can be precisely controlled by applying electric and magnetic fields or mechanical shear stresses. To exploit this capability, this study proposes a nonchemical, structure-driven approach to enhance the intrinsic energy dissipation performance of polymer-graphene nanocomposites. This finding underscores the critical role of the nanostructural design of fillers in tailoring the damping performance of composite systems by identifying the physicochemical factors that determine their interfacial properties.

4. Conclusion

In this study, we provide theoretical insights into the influence of graphene chirality on the energy dissipation efficiency of polymer nanocomposites. The physicochemical factors governing the damping behavior of the epoxy/graphene systems were systematically characterized using a computational approach.

The results show that the chirality-induced anisotropy of graphene is an important factor in the damping performance of the entire nanocomposite domain. For the shear strain along the axis direction of the armchair graphene, the load-bearing capacity of the entire system was significantly reduced compared to the perpendicular condition (i.e., along the axis direction of the zigzag graphene). This trend was also observed in the vibration tests in which the unidirectional shear deformation was repeated in a reciprocating motion. The accumulated residual strain due to repeated loading was more pronounced in the armchair graphene than in the zigzag graphene. Analysis of the spatial characteristics within the microstructure clearly revealed that anisotropic energy dissipation was caused by an interfacial mismatch and the associated slippage behavior. In addition, the displacement of the polymer matrix around the graphene also increased under the load in the armchair direction compared to the load in the zigzag direction. This indicates that the mechanical bond was weaker and the frictional energy dissipation at the interface was amplified under the armchair load.

Two principal mechanisms were proposed to explain this anisotropic behavior. First, the intrinsic elastic modulus of graphene is higher along the armchair direction. This suppressed the out-of-plane roughness and maintained the planar integrity of the nanofiller during mechanical loading. Consequently, interfacial wrinkling and mechanical interlocking with the surrounding polymer chains were inhibited, facilitating sliding and enhanced energy dissipation. Second, the surface energy landscape of graphene varies with the chirality. Along the armchair axis, the elevated energy barriers forced the polymer chains to bypass high-energy sites, producing longer molecular trajectories. Conversely, zigzag-oriented interfaces enable polymer chains to follow low-energy valleys, thereby preserving interfacial adhesion and reducing frictional

energy dissipation.

This study offers molecular-level insight into a structural, nonchemical strategy for enhancing the intrinsic energy dissipation capability of polymer-graphene nanocomposites. This strategy is particularly suitable and scalable in applications such as area-selective cancellation of ultrahigh-frequency vibrations, design of metamaterials considering the anisotropy of nanofillers, and advanced interface mechanics modeling of graphene-embedded nanocomposites.

Although this study focused solely on pure graphene, functionalized graphene is expected to be a promising research direction in the near future. By introducing functional groups, the graphene–polymer interface can be roughened, thereby enhancing interfacial friction and promoting more efficient load transfer. On the other hand, excessive functionalization may disturb physical interlocking and weaken the effectiveness of load transfer, suggesting a possible strengthening–weakening crossover in the damping ratio depending on the degree of functionalization. Functional groups also can increase surface polarity and disrupt the planar graphene lattice, which further alters the mechanisms of energy dissipation. Because the actual impact strongly depends on the type, density, and distribution of functionalization, a systematic investigation under controlled conditions will be necessary. Therefore, elucidating the role of functionalization in controlling interfacial slip and load transfer is an important next step in this research.

CRediT authorship contribution statement

Sihyun Kim: Writing – original draft, Visualization, Software, Methodology, Data curation. **Hongdeok Kim:** Writing – review & editing, Visualization, Software, Methodology. **Junho Oh:** Supervision, Project administration. **Joonmyung Choi:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.surfin.2025.107763.

Data availability

Data will be made available on request.

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